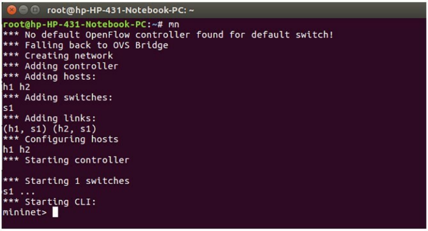

Figure 1: Mininet in action

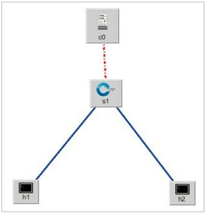

Figure 2: Topology created using Miniedit (an open source tool to create Mininet topologies)

Figure 3: Nodes and net

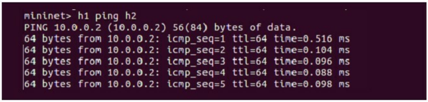

Figure 4: The command makes *h1* ping *h2*

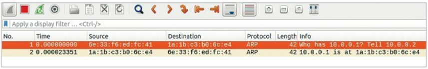

Figure 5: Packet captured at the interface *s1-eth1.*

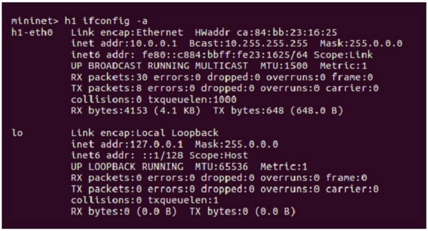

Figure 6: Getting details of host1 interface

The above command will list all the nodes present in the created network. As shown in Figure 3, the nodes s1, h1, h2 are displayed.

The command to display and list the links present in the network is:

```
Mininet>net
```

As shown in Figure 3, the interface *eth0* of the host *h1* is connected to *eth1* of switch *s1* and the interface *eth0* of host *h2* is connected to *eth2* of switch *s2*.

The command to display the IP addresses and the process IDs of the nodes is:

```
Mininet>dump
```

As seen in Figure 3, *h1* is assigned the IP address 10.0.0.1 with the process ID 4403 and *h2* is assigned the IP address 10.0.0.2 with the process ID 4407.

- The command to ping a specific host to a targeted host is:

```
Mininet> h1 ping h2
```

On inspecting the packets, we can see that the first ping has taken considerably longer (0.805) than the others. This is because ARP tables, MAC tables, etc, are initialised during the first ping.

The command to display the address information of the nodes is:

```
Mininet> h1 ifconfig -a
```

This command will display the IP address, broadcast address and MAC address of the host *h1*, as in Figure 6.

- The command to test the connectivity among hosts is:

```
Mininet>pingall
```